Prodigy Anesthesia Quick Review

Chronic Pain

What is radiculopathy?	Radiculopathy is a functional abnormality associated with at least one nerve root *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is phantom pain?	Phantom pain is the sensation of pain in a limb that has been amputated (the phantom limb). *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1658.*
What is neuropathic pain?	Neuropathic pain involves a complex interaction between peripheral and central pain mechanisms that are usually associated with lesions of peripheral nerves, nerve roots, ganglions, or spinal structures. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1027.*
What is allodynia?	Allodynia is the perception of an ordinarily non-painful stimulus as pain *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is the definition of chronic pain?	Chronic pain is any pain that persists longer than usual for an acute disease or longer than the reasonable time in which the condition should normally be able to heal. This is most likely to be within 1 and 6 months. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1027.*
What is paresthesia?	A paresthesia is any abnormal sensation (numbness, tingling, pins and needles sensation) that occurs without any stimuli *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is neuralgia?	Neuralgia is pain that follows the distribution of a nerve or group of nerves *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is hypoesthesia?	Hypoesthesia is a reduced ability to sense cutaneous stimuli such as light touch, pressure, or temperature. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*

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What is hyperpathia?	Hyperpathia is a combined disorder consisting of hyperesthesia, allodynia, and hyperalgesia. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is hyperalgesia?	Hyperalgesia is an increased or exaggerated response to painful stimuli *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is hypoalgesia?	Hypoalgesia is a diminished response to painful stimuli *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is dyesthesia?	Dyesthesia is the presence of an unpleasant sensation whether or not a causative stimulus is present. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is anesthesia dolorosa?	Anesthesia dolorosa is the perception of pain in an area that lacks sensation. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is anesthesia?	Anesthesia is defined as the lack of all sensation, painful or otherwise. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is hyperesthesia?	Hyperesthesia is an exaggerated response to a mild stimuli. This is in contrast to hyperalgesia, which is an exaggerated response to a painful stimuli. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*
What is analgesia?	Analgesia is the lack of pain perception *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.*

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What are the two types of Complex Regional Pain Syndrome and what were their former names?	They are identified as CRPS I and CRPS II. CRPS I was originally referred to as reflex sympathetic dystrophy. CRPS II was originally referred to as causalgia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
What are the symptoms of complex regional pain syndrome?	Pain that occurs spontaneously without an apparent stimulus, hyperalgesia, allodynia, and sudomotor and vasomotor dysfunction. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
What settings are typically used on a conventional transcutaneous electrical nerve stimulations (TENS) unit when treating chronic pain?	Conventional TENS utilizes a current of 10-30 milliamps, a pulse width of 50-80 microseconds, and a frequency of 80-100 Hz. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1081.*
How are CRPS I and CRPS II treated?	The primary modes of treatment for CRPS focus on sympathetic blocks, physical therapy, and oral medications such as gabapentin and memantine. IV regional anesthesia has also been employed for the treatment of CRPS. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1658.*
What is the primary difference between CRPS I and CRPS II?	They present with the same symptoms and clinical characteristics except that CRPS II (formerly known as causalgia) is preceded by trauma to a nerve. CRPS II is usually caused by high velocity injuries to large nerves (most commonly the brachial plexus) and has an immediate onset. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
What are the treatment options for phantom limb pain?	Phantom limb pain is treated with oral medications such as gabapentin and antidepressants. Transcutaneous electrical nerve stimulation, spinal cord stimulators, and biofeedback have also been used in the treatment of phantom limb pain. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1658.*
What are the complications associated with epidural steroid injection?	Needle trauma, vasospasm, and infection. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1652.*

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What type of injection would be most likely to improve symptoms in a patient with facet syndrome?	Medial branch blocks and facet joint injections are efficacious in patients with facet syndrome. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1653.*
What are the potential complications from transforaminal steroid injection and how do they occur?	Injury to the brain or spinal cord can occur. Trauma to the brain is associated with damage to the vertebral artery, vasospasm, or a particulate embolus via inadvertant injection into the vertebral artery. Injury to the spinal cord can occur via similar mechanisms that involve the radicular artery adjacent to the nerve root. In the lumbar area, anatomical variants of the artery of Adamkiewicz also present a potential location of inadvertant injection or trauma. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1652.*
How long would you expect that a radiofrequency rhizotomy of a medial branch would provide relief of symptoms in a patient with facet syndrome?	3-12 months *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1654.*
What are the characteristics of facet syndrome?	Facet syndrome typically results in pain that begins in the lower back and radiates through the posterior thigh and ends at the knee. Paraspinal tenderness is usually present on physical examination and the pain is reproduced with extension and rotation movements of the lower back. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1653.*
When would a radiofrequency rhizotomy be indicated for a patient with facet syndrome?	Results from a medial branch block can have a relatively long duration (3-6 months). If the results from a medial branch block are short-lived, a radiofrequency rhizotomy should be *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1653-1654.*
With what forms of chronic pain is a neuroendocrine stress response generally associated?	A neuroendocrine stress response is typically only seen in syndromes associated with severe recurring pain related to peripheral mechanisms or significant central pain mechanisms such as those associated with paraplegia. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1038-1039.*

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What are the central mechanisms that contribute to neuropathic pain?	Central mechanisms include the reorganization of neural connections, wind-up of wide dynamic range neurons, spontaneous neural discharges, and loss of inhibitory mechanisms in neural segments. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1037.*
What are the peripheral mechanisms that contribute to neuropathic pain?	Receptor sensitization to chemical, thermal, and mechanical stimuli, spontaneous neural discharge, and up-regulation of adrenergic receptors. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1037.*
What is the chief danger in performing cervical transforaminal steroid injections?	The path of the needle for these blocks has been demonstrated to be within 2 millimeters of the vertebral, ascending cervical, and deep cervical artery making intra-arterial injection or vascular trauma a significant risk. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 652.*
How long do epidural steroid injections relieve symptoms? What are the implications of this?	Studies have demonstrated that the efficacy of epidural steroid injections lasts no more than 3 months. Because of this, the American Academy of Neurology recommends against the routine use of epidural steroid injections. The argument for their use by other clinicians is that epidural steroids can be used as a modality to reduce the need for opioids or potent anti-inflammatory drugs during the acute phase of injury. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1651.*
What are the treatment options for pyriformis syndrome?	Physical therapy combined with nonsteroidal anti-inflammatory and muscle relaxant drugs and local anesthetic/steroid injection directly into the pyriformis muscle. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1655.*
How do epidural steroids relieve symptoms from a herniated nucleosus pulposus?	When the disc ruptures, it releases cytokines and inflammatory mediators that cause a chemical inflammation of the adjacent nerves. One of the mediators released in high quantities is phospholipase A2. Epidural steroids can reduce inflammation by inhibiting the activity of phospholipase A2. Epidural steroids also have a local anesthetic quality and exert an antinociceptive effect. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1651.*

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What is the fundamental pathology that is believed to underlie the allodynia and hyperalgesia associated with persistent painful stimuli?	The concept of wind-up (also called central sensitization) underlies these phenomena. Chronic C fiber stimulation can enhance the response to subsequent stimulation and result in neuronal excitability. As the cycle progresses, low level, nonpainful stimuli can begin to produce pain. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1649-1650.*
How many epidural steroid injections may be performed if there is no response to the first injection?	Up to three injections may be performed, but multiple injections should not be administered within a short period of time. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1652.*
What are the possible complications from a celiac plexus block?	Abdominal aortic dissection, paraplegia, orthostatic hypotension, retroperitoneal hematoma, hematuria, pleurisy, and transient motor paralysis. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1659.*
How do antidepressants help control chronic pain?	Antidepressants provide analgesia by inhibiting the presynaptic reuptake of norepinephrine, serotonin, or both. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1055.*
What spinal pathway transmits the discriminative features of pain such as location, intensity, and duration?	The axons of most second-order neurons cross to the opposite side of the spinal cord before forming the spinothalamic tract. The spinothalamic tract is divided into lateral and medial tracts. The lateral spinothalamic tract carries discriminatory features of pain such as location, intensity, and duration while the medial tract sends fibers to the thalamus which modulates the autonomic and emotional perceptions of pain. The lateral spinothalamic tract may also be referred to as the neospinothalamic tract. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1031-1032.*
What are wide-dynamic range neurons and in which of Rexed's lamina are wide-dynamic range neurons most abundant?	Wide dynamic range neurons are the most abundant nerve fiber type in the dorsal horn and are found primarily in lamina V. They carry noxious stimuli in addition to receiving input from A-beta, A-delta, and C fibers. During repeated stimulation, wide-dynamic range neurons increase their firing rate exponentially despite no change in stimulus intensity. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1030.*

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What nerves carry pain fibers from the head?	Pain fibers from the head are carried within the trigeminal, facial, glossopharyngeal, and vagus nerves. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1028.*
Where are the primary afferent neurons located?	The primary afferent neurons are found within the dorsal root ganglia. The dorsal root ganglia are located within the vertebral foramina at each spinal cord level. An easy way to remember the location of the sensory neurons is to remember SAD, or Sensory Afferent Dorsal, meaning that the sensory neurons are afferent and are located in the dorsal ganglion. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1028.*
What are the principal neurotransmitters in the activation of dorsal horn neurons following painful stimuli?	The principal neurotransmitters responsible for activation of dorsal horn neurons in the transmission of noxious stimuli are glutamate and substance P, although other neuropeptides such as calcitonin, gene-related peptide, adenosine triphosphate, somatostatin, and growth factors have also been implicated. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1647-1648.*
How does the stimulation of alpha-2 adrenergic receptors in the spinal cord influence pain sensation?	Stimulation of alpha-2 adrenergic receptors in the spine can produce analgesia by inhibiting nociceptive afferent fibers. It has also been shown to produce muscle flaccidity in animal studies by hyperpolarizing muscle neurons. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 920.*
What substances are capable of sensitizing mechanoheat receptors to painful stimuli?	Histamine, bradykinin, substance P, prostaglandins, leukotrienes, and serotonin are all capable of sensitizing A-delta and C mechanoheat nociceptors to pain. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1033.*
Which nerve conduction pathway is most likely to be associated with the autonomic and emotional responses to pain?	The paleospinothalamic tract (medial spinothalamic tract) projects to the medial thalamic nuclei and is associated with autonomic and emotional responses to pain. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1031.*

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At what spinal level should a cervical epidural steroid injection be made?	Because the C6-C7 and C7-T1 interspaces have the largest interlaminar distances, the injection is often made at this point regardless of the level of the pathology and the practitioner relies on the flow of drug to the level of pathology. This technique can be used in all interspaces below C3-C4. Interlaminar injections at the C2-C3 interspace have not been described in the literature. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1576.*
At what spinal level is a lumbar sympathetic block performed?	The sympathetic nerves to the lower extremities leave the spinal cord above the L2 level and pass through a 'gateway' sympathetic ganglion at the L2 level. A lumbar sympathetic block can be achieved by performing a single local anesthetic injection just below that at the L3 level. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1075.*
What are the indications for a lumbar sympathetic block?	A lumbar sympathetic block may be performed for the treatment of chronic pelvic pain or peripheral vascular disease. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1074.*
What is pyriformis syndrome?	Pyriformis syndrome originates as pain in the buttocks. It can occur due to trauma, surgery, infection, or an anatomical abnormality in which the pyriformis muscle is bifurcated and a branch of the sciatic nerve travels within one of the abnormally split pyriformis bundles. The pain usually radiates from the greater trochanter to the ipsilateral buttock. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1655.*
What drugs are generally administered to perform a celiac plexus neurolysis?	Either phenol or alcohol are used in a neurolytic celiac plexus block. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1658-1659.*
What are the clinical characteristics of sacroiliac disease?	Patients with pyriformis syndrome exhibit pain in the area of the sacroiliac joint with radiation to the groin, medial buttocks, and posterior thigh with occasional radiation below the level of the knee. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1655.*

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A celiac plexus block would not relieve pain arising from what abdominal stuctures?	The left colon and the pelvic viscera *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1658.*
For what conditions would a celiac plexus block be indicated?	A celiac plexus block is most commonly performed for cancer of the upper abdomen, including the liver, pancreas, gallbladder, stomach, spleen, kidneys, intestines, and adrenal glands. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1658-1659.*
When does phantom pain first occur?	It may present immediately, but phantom pain usually occurs within a few days of the amputation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1658.*
What are the chronic pain-related symptoms that develop from diabetic neuropathy?	The primary symptoms are an unpleasant tingling, burning, distal sensory loss, and aching in the lower extremities. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 271.*
What percentage of insulin-dependent diabetics exhibit neuropathies?	65% *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
When is the optimal time to perform an epidural block for the treatment of postherpetic neuralgia?	It should be performed within 2-4 weeks of the onset of the herpes zoster rash to have the greatest effect. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
What are the treatment options for postherpetic neuralgia?	Antiviral drugs (acyclovir, famcyclovir, or valacyclovir) administered during the episode of herpes zoster reduce the duration and severity of the herpes zoster outbreak and decrease the incidence of conversion to postherpetic neuralgia. Epidural blocks have been shown to treat pain in many of these patients. Anticonvulsants, antidepressants, and opioids are also used in the medical management of postherpetic neuralgia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
What is the incidence of postherpetic neuralgia in patients with herpes zoster?	About 10-15% of patients who suffer from herpes zoster will develop postherpetic neuralgia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*

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What is postherpetic neuralgia?	Postherpetic neuralgia is pain that persists more than three months after resolution of the rash associated with herpes zoster. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
What pain pathway is associated with nondiscriminatory pain sensations?	The spinomesencephalic tract projects to the midbrain reticular formation and may generate nondiscriminatory pain sensations. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1031-1032.*
What activities aggravate pyriformis syndrome?	Prolonged sitting and getting up from a seated position. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1655.*
What specific nerve block can relieve symptoms from sacroiliac joint syndrome?	Blockade of the medial branch of the dorsal rami of L5 and S1-S3 can relieve pain from sacroiliac joint syndrome. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1657.*
What are the treatments available for sacroiliac joint syndrome?	Physical therapy, SI joint injection, radiofrequency denervation, and surgical fusion of the SI joint. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1655.*
What is tic doloureux?	Tic doloureux (trigeminal neuralgia) is a painful condition involving the trigeminal nerve (usually the V2 or V3 branches). Like other forms of neuropathic pain such as shingles, reflex sympathetic dystrophy, phantom limb pain, and central postherpetic pain, it results from abnormalities in or damage to the nervous system. *Lee CY. Manual of Anesthesiology. Singapore: McGraw-Hill; 2006: 361, 408-409.*

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